

Krishnaa Vadivel | PhD Research Engineer | Scientific Computing | ML | EE / Embedded Systems

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Summary

Electrical engineering PhD researcher and engineer with experience in first-principles materials modeling, semiconductor-device simulation, scientific ML, embedded systems, and performance-oriented software. Combines theory and implementation across Python/C++, CUDA/ROCm, FPGA/MCU platforms, and leadership-class HPC environments.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Develop first-principles and many-body workflows for self-trapped excitons, exciton-phonon coupling, semiconductor-relevant modeling, and reduced effective theories for materials research.
- Build distributed scientific-computing infrastructure in Python and C++ using MPI, OpenMP, PETSc, SLEPc, CUDA, and ROCm for Frontier, Perlmutter, and related HPC systems.
- Investigate ML-assisted acceleration of scientific workflows, including equivariant graph neural networks, molecular-property prediction, and agentic tooling for documentation- and paper-driven input generation.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and delivered APS presentations in 2024, 2025, and 2026 on self-trapped exciton and exciton-polaron research.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based tools for waveform processing, visualization, and engineering diagnostics in deployed geological well-logging systems.
- Implemented embedded and digital functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms; supported circuit bring-up, oscilloscope validation, and targeted Altium PCB updates.
- Contributed host-side tooling with C#, ASP-style services, and Microsoft SQL-backed workflows for engineering and field-data operations.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content on lasers, optics, and photonics devices for engineering-facing audiences.

Selected Projects

Scientific HPC Research: Distributed first-principles and many-body workflows for exciton-phonon problems, materials simulation, and reusable accelerator-enabled compute pipelines.

Semiconductor Device Modeling: Numerical treatments of Poisson, drift-diffusion, and pn-junction-style equations bridging physics, EE, and computational modeling.

Scientific ML and Automation: Equivariant molecular ML, agentic RAG workflows, and structured tooling for documentation-driven scientific input generation.

Embedded and Digital Systems: FPGA/MCU-based engineering work spanning sensor logging, a published surface-EMG robotic hand, digital design, and hardware-software integration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Programming: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Scientific / ML: PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas, SciPy, SymPy, matplotlib

Distributed Compute: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, ROCm

Hardware / Systems: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, Linux, Docker, Kubernetes, gcloud

Domains: Computational physics, semiconductor devices, photonics, embedded systems, scientific software